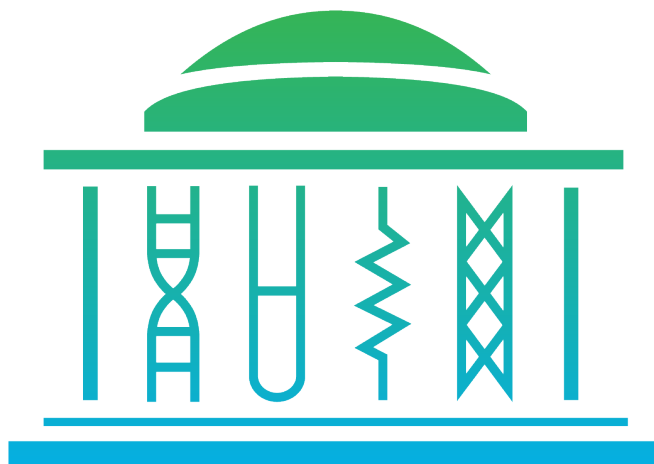


Science Olympiad
Chemistry Lab Test
MIT Science Olympiad



Team Name: _____

Team Number: _____

Directions:

- This test has three sections: multiple choice and two sections corresponding to the rotating topics. The multiple choice questions are worth half of the total points, while each topic sections is worth a quarter of the total points.
- One of the topics will be a lab question; we will provide a list of the equipment and chemicals you will need to complete the lab. Once the test begins, please check that you have everything listed.
- Please keep your goggles and lab coat/apron on at all times, even if you are not working on the lab practical!
- You are not expected to complete the entire test; feel free to skip around and answer the questions that you can.
- For short answer questions, up to half credit may be given for correctly shown work or calculations with an incorrect final answer.
- Once time is called, please put down your pencils immediately and stop working.
- In the event of a tie, we will compare scores on the *last* problem; if still tied, compare the second-to-last; continue backward until the score differs.
- Good luck and have fun!

Multiple Choice

- (2 points) What type of reaction is the reaction: $\text{Zn} + \text{S} \longrightarrow \text{ZnS}$?
 - Synthesis
 - Decomposition
 - Comproportionation
 - Single displacement
- (2 points) A student performs the reaction $2\text{HI} + \text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{I}_2$. Which of the following correctly relates the rate of change of the reactants and products?
 - $\frac{d[\text{HI}]}{dt} = -\frac{d[\text{H}_2\text{O}]}{dt}$
 - $\frac{d[\text{HI}]}{dt} = -2\frac{d[\text{H}_2\text{O}]}{dt}$
 - $\frac{d[\text{HI}]}{dt} = -2\frac{d[\text{H}_2\text{O}]}{dt} + \frac{d[\text{I}_2]}{dt}$
 - None or multiple of the above.
- (2 points) A reaction $3\text{A} + \text{B} \longrightarrow 3\text{C} + \text{D}$ has the rate law $R = k[\text{A}][\text{B}]^3[\text{D}]^{-1}$. If the initial concentration of A is doubled and the initial concentration of D is halved, what is the expected change in initial rate of the reaction?
 - There is no change in initial rate
 - The initial rate will double
 - The initial rate will quadruple
 - The initial rate will be multiplied by 16
- (2 points) Chemical A is able to irreversibly turn into chemicals B and C through different pathways. A sample of A turns into 25% B and 75% C at STP without a catalyst. Is the addition of a catalyst able to change these proportions?
 - No, a catalyst cannot change the proportion of products and must speed up the conversion into both by the same factor.
 - No, a catalyst will convert B and C back into A.
 - Yes, a catalyst can speed up the formation of one more than the other.
 - Yes, a catalyst will increase the rate of temperature increase, which can in turn affect the proportions of B and C produced.
- (2 points) Which of the following reactions is correctly balanced?
 - $2\text{HI} + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{I}_2$
 - $2\text{CuCO}_3 + \text{H}_2\text{O} \longrightarrow \text{CO}_2 + \text{Cu}_2(\text{CO}_3)(\text{OH})_2$
 - $\text{C}_6\text{H}_6 + \text{I}_2 \longrightarrow \text{C}_6\text{H}_5\text{I} + \text{HI}$
 - $3\text{Br}_2 + 2\text{I}_2 \longrightarrow 2\text{IBr}_3$

6. (2 points) Bob wants to make piranha solution (don't do this irl kids u will die), and slowly mixes 5.0 mls of 12.0M H_2O_2 into 10.0 mls of 18.0M H_2SO_4 while the solution is on ice. What is the concentration of H_2O_2 in the final mixture if the final volume is 14.5 mls?
- A. 12.41M
 - B. 6.21M
 - C. 4.00M
 - D. 4.14M
7. (2 points) Sid is performing an acid base titration with solely a solution of citric acid, a triprotic acid, and a solution of sodium hydroxide, NaOH. To neutralize half of the citric acid in 20 mls of a 2M solution of citric acid, how many mls of a 1M solution of NaOH are needed?
- A. 30 mls
 - B. 60 mls
 - C. 120 mls
 - D. 10 mls
8. (2 points) When $\text{C}_2\text{H}_6\text{O}$ is passed through hot CuO , the products are $\text{C}_2\text{H}_4\text{O}$, Cu_2O , and H_2O . Give the sum of the coefficients of all chemicals in this reaction when balanced with the smallest integer coefficients.
- A. 5
 - B. 6
 - C. 7
 - D. 8
9. (2 points) Activated carbon has a very high surface area, and thus is able to adsorb many chemicals, including iodine. Briefly, the iodine number essentially describes how many milligrams of iodine can be absorbed per gram of activated carbon. Bob obtains a sample of activated carbon with an iodine number of 600 mg/g. How many moles of elemental iodine (I_2) can be absorbed by 32.2 grams of Bob's activated carbon?
- A. 0.0190 mols
 - B. 0.0381 mols
 - C. 0.0761 mols
 - D. 1.61 mols
10. (2 points) The reaction $\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ has what sign in its enthalpy at 25C? Does the reaction speed up or slow down with a 10 degree celcius increase in temperature?
- A. Positive, Slow down
 - B. Zero, Cannot be determined
 - C. Negative, Speed up
 - D. Cannot be determined, Speed up

11. (2 points) Which graph for a second order reaction in $[A]$ is linear?
- $\ln([A])$ vs $-\ln(\text{time})$
 - $[A]$ vs $1/\text{time}$
 - $1/[A]^2$ vs time^2
 - $1/[A]$ vs time
12. (2 points) The rate of the irreversible reaction, $\text{CH}_3\text{I} + \text{OH}^- \longrightarrow \text{I}^- + \text{CH}_3\text{OH}$ is monitored, and the following data is observed. Determine the closest integer reaction order in CH_3I and OH^- .

Reaction	$[\text{CH}_3\text{I}]$	$[\text{OH}^-]$	$[\text{I}^-]$	$[\text{CH}_3\text{OH}]$	Initial Rate
1	1.10M	0.23M	0.10M	20.0M	$1.74 \times 10^{-3} \text{ M/s}$
2	2.21M	0.23M	0.10M	20.0M	$3.43 \times 10^{-3} \text{ M/s}$
3	3.37M	0.22M	0.20M	20.0M	$5.27 \times 10^{-3} \text{ M/s}$
4	7.78M	0.44M	1.30M	20.0M	$2.41 \times 10^{-2} \text{ M/s}$

- 0th order for both.
 - 1st order in OH^- , 0th order in CH_3I
 - 2nd order in CH_3I , 1st order in OH^-
 - 1st order for both.
13. (2 points) It is known that chemically, the reaction order of CH_3OH is zero. However, the inclusion of CH_3OH is still important in this experiment. Explain why this may be the case.
- CH_3OH is present in the reaction equation, and its inclusion is necessary for the reaction to occur.
 - CH_3OH makes the mixture less toxic and reduces fumes, aiding to any environmental and safety concerns.
 - CH_3OH binds to I^- , preventing the reverse reaction from occurring.
 - CH_3OH increases the solubility of CH_3I , allowing for high concentrations to be used in the experiment.
14. (2 points) A student passes a stream of dilute acetone vapor over a small, 1 gram piece of heated copper wire in a tube, which catalyzes the oxidation of acetone by oxygen into various products. Which of the following will most likely increase the percentage of acetone oxidized?
- Use 2 grams of identical copper wool to increase the surface area.
 - Pass acetone liquid instead of acetone gas over the hot copper, as more moles of acetone will be passed through the tube.
 - Use a piece of heated zinc, as zinc is more susceptible to oxidation by H_3O^+ than copper.
 - Decrease the gauge (increase the diameter) of the copper wire while keeping the mass constant, as this is less costly.
15. (2 points) Balance the equation $\text{I}_2 + \text{NH}_3 \longrightarrow \text{NI}_3 \cdot \text{NH}_3 + \text{NH}_4\text{I}$, then give the sum of the coefficients of the reactants only.

- A. 5
B. 8
C. 12
D. 15
16. (2 points) Calder carelessly leaves a 1.000 gram sample of cesium in completely dry air, and miraculously it has not spontaneously burst into flames. When he returns, all of the metallic cesium has oxidized in air to form a mixture of oxides (Cs_2O), peroxides (Cs_2O_2), and superoxides (CsO_2). Luckily, he left the cesium on a weighing dish, and found that the mass had increased by 0.241 grams. Is it possible to determine the percentage of cesium oxide present in the sample based on just this data?
- A. No, it is not possible to distinguish between cesium peroxide and a mix of cesium oxide and superoxide.
B. No, the decrease in density of the mixture significantly impacts the percentage, and density data is necessary.
C. Yes, the data suggests that there is zero cesium oxide.
D. Yes, the data suggests that it is approximately 200% cesium oxide by mass.
17. (2 points) Balance the reaction that occurs when magnesium metal reacts with dilute nitric acid to form nitrous oxide, N_2O . Then, give the mass of N_2O that can be formed using 3 grams of magnesium, 10 mls of 6M nitric acid, 30 mls of 12M H_2SO_4 , and 100 mls of water, if the reaction has a yield of 70% with respect to the limiting reagent.
- A. 0.924 grams
B. 1.848 grams
C. 1.904 grams
D. 2.64 grams
18. (2 points) Bob, liking to destroy things, burns 10.0 grams of $\text{C}_6\text{H}_5\text{N}$ in excess oxygen to completion. He notes that the final mixture of gasses is colorless. What is the composition of the final gas mixture (excluding oxygen), and how much of the nitrogen-containing compound is formed?
- A. CO_2 , H_2O , N_2 , 0.77 grams
B. CO_2 , H_2O , N_2 , 1.54 grams
C. CO_2 , H_2O , NO , 3.29 grams
D. CO_2 , H_2O , NO_2 , 5.05 grams
19. (2 points) Bill complains that the problems above are too boring, so his teacher tells him to go do another titration as a punishment. This time, he is to titrate a sample of 2.000 grams of benzoic acid contaminated with naphthalene (an unreactive solid). Which of the following is the best choice for a titrant? He has ~50 mls of each of the following titrants.
- A. 1.00M Lithium hydroxide
B. 18.000M Sodium hydroxide

- C. 1.00M Sodium acetate
D. 1.00M Sodium benzoate
20. (2 points) For some reason, a random biology lab is requesting deuterated sodium hydroxide (NaOD). Determine how many grams of deuterated sodium hydroxide can be made by combining 39.02 grams of sodium oxide with 11.00 grams of D_2O .
- A. 45.11 grams
B. 50.12 grams
C. 51.60 grams
D. 54.12 grams
21. (2 points) Balance the following reactions with the smallest possible integer coefficients, then give the sum of the coefficients of OH^- of the three reactions.
- Reaction 1. $\text{Ca} + \text{H}_2\text{O} \longrightarrow \text{Ca}^{2+} + \text{OH}^- + \text{H}_2$
 - Reaction 2. $\text{OH}^- + \text{Br}_2 \longrightarrow \text{BrO}_3^- + \text{Br}^- + \text{H}_2\text{O}$
 - Reaction 3. $\text{Zn}^{2+} + \text{OH}^- \longrightarrow \text{Zn}(\text{OH})_4^{2-}$
- A. 5
B. 6
C. 11
D. 12
22. (2 points) Isaac has really gotten into titrations, and titrates his sample of his favorite rock, a mixture of CaC_2O_4 and $\text{Ca}_3(\text{PO}_4)_2$, with his favorite oxidizing agent, potassium dichromate. He grabs 1.42 grams of this stone, grinds it up finely, and adds 30 mls of warm 3M sulfuric acid while stirring for half an hour, producing $\text{H}_2\text{C}_2\text{O}_4$ and H_3PO_4 in solution. After the solution cools down, he filters out any solids using a glass fritted buchner funnel and dilutes the solution to 100.00 mls. He finds that he needs just 2.03 mls of his 0.1M dichromate solution to neutralize a 10.0 mls aliquot of this solution. Determine the percentage of CaC_2O_4 by mass of his stone.
- A. 44.9%
B. 54.9%
C. 24.48%
D. He must have done something wrong, as the mass % is above 100%
23. (2 points) When Jeffery repeats Isaac's experiment, he is impatient, and uses 30 mls of hot 18M sulfuric acid to process his stones. Explain how this would affect the determined percentage mass of CaC_2O_4 in his stones.
- A. His calculated mass would be lower than the actual mass because some of the oxalic acid would decompose in the hot acid.
B. His calculated mass would be lower than the actual mass, because the dichromate would react with the concentrated sulfuric acid to form CrO_3 .

- C. His calculated mass would be higher than the actual mass, because he would extract too much of the oxalate from the stones.
- D. His calculated mass would be either higher or the same as the actual mass, because the density of 18M sulfuric acid is higher than that of 3M sulfuric acid, even when hot.
24. (2 points) One day, a 50.00 gram rod of Co-60 magically appears on Séamus's desk. Unfortunately, he has forgotten what he needs to do in such a situation, and immediately scrolls Twitter for 30 minutes to determine what he should do, after which he runs straight to the ER. Given that the half life of Co-60 is 5.72 years, determine the approximate number of atoms of Co-60 decayed in those 30 minutes of scrolling Twitter. Approximate 365 days = 1 year and molar mass of Co-60 to be 60 g/mol.
- A. 6.0×10^{18} atoms
- B. 5.5×10^{18} atoms
- C. 5.0×10^{19} atoms
- D. 3.5×10^{18} atoms
25. (2 points) Merricks is trying to impress his father, so one day he attempts to build a flamethrower. Unfortunately, he is not very well versed in flamethrower-building, and tries to fuel it using a mixture of dinitromethane, dimethylsulfoxide (DMSO), and pentane ($\text{CH}_2\text{N}_2\text{O}_4$, $\text{C}_2\text{H}_6\text{OS}$, C_5H_{12} , respectively). The flamethrower consumes fuel at a rate of 17.2 grams per second, and it successfully runs for exactly 3 seconds before malfunctioning and exploding. The only nitrogen-containing compound produced in the 3 seconds is 9.66 grams of N_2 , and the only sulfur-containing product is 2.39 grams of SO_2 . If the fuel burns to completion, how many grams of CO_2 were produced?
- A. 14.17 grams
- B. 43.12 grams
- C. 55.73 grams
- D. 134.44 grams

Ancient Life on Mars

In 2025, NASA's Perseverance rover reported that a crater-rim rock sample at Jezero Crater is made of serpentine-rich minerals—evidence for extensive water–rock reaction in Mars' ancient crust. At the same time, peer-reviewed rover geochemistry and mineralogy provided in situ evidence for serpentinization within igneous rocks of Jezero's crater floor (Mááz formation), linking these alteration textures and mineral assemblages to H_2 production. A second 2025 study described carbonated ultramafic rocks encountered along Perseverance's traverse—coarse olivine plus Mg/Fe-carbonates and alteration minerals—consistent with exposure to both liquid water and CO_2 , recording conditions that can preserve ancient environmental chemistry in minerals.

The process of serpentinization is the hydration of olivine-rich igneous rocks. In simplified terms, olivine reacts with water to form new minerals and produce a strongly reducing solution containing Fe^{2+} ions that can release H_2 via reduction of water. The resultant H_2 can go on to reduce CO_2 into CH_4 and other reduced carbon species.

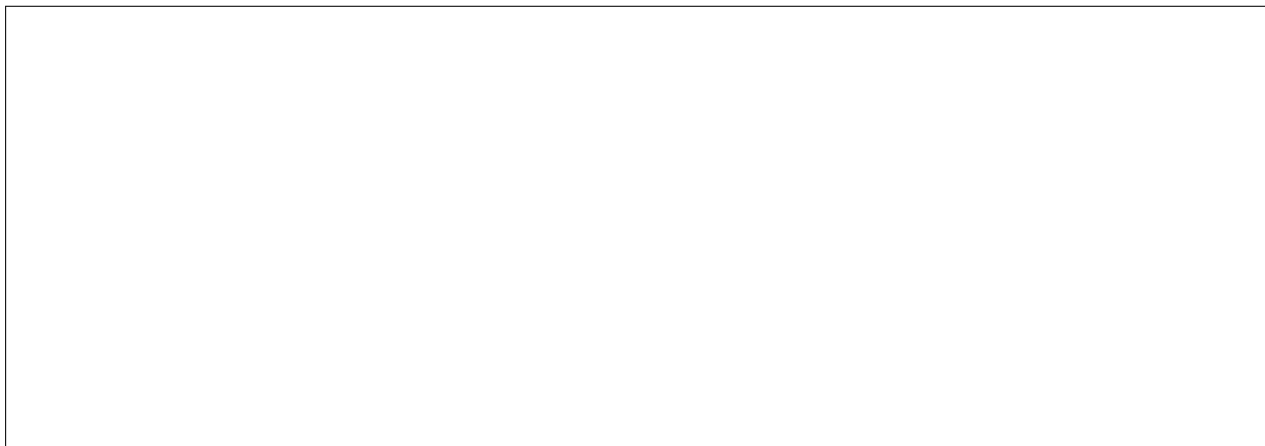
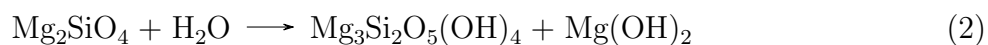
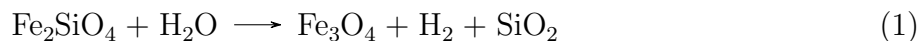
In the context of astrobiology, this reducing environment is very important: On Earth, rich, alkaline water–rock systems can supply chemical energy for chemosynthetic metabolisms; on Mars, the combination of serpentine-rich rocks suggesting presence of water in the past and evidence for H_2 -producing alteration makes serpentinized environments high-priority targets for assessing habitability and for searching for preserved biosignatures.

26. (2 points) Olivine is a solid solution series of forsterite, Mg_2SiO_4 , and fayalite, Fe_2SiO_4 . As a result, olivine has the non-stoichiometric chemical formula $\text{Mg}_{2k}\text{Fe}_{2-2k}\text{SiO}_4$, where k is a value ranging from 0 to 1. For this question, we assume $k = 0.9$. Calculate the mass percent of Si in this olivine.

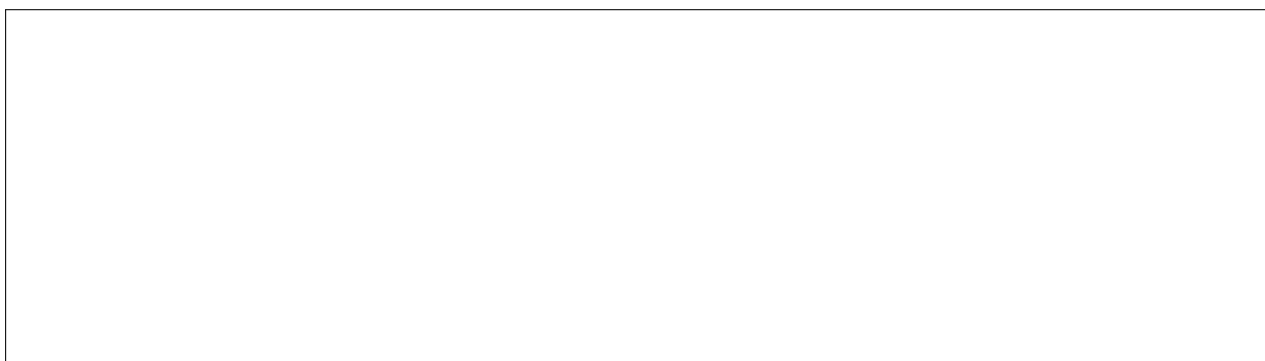
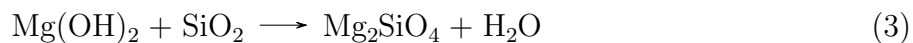
27. (3 points) It is difficult to work with non-integral (or even worse, variable) coefficients when balancing reactions, so we'll take a different approach to analyzing the reactions involving

olivine.

The key concept is that linear combinations of balanced reactions will still be balanced. We'll treat olivine as a mixture of Mg_2SiO_4 and Fe_2SiO_4 . Balance the following two reactions.



28. (3 points) Note that silica SiO_2 appears on the right-hand side of the hydrolysis of fayalite. Balance both of the reactions below. You can use a linear combination of the balanced versions of reaction (2) and (3) to find (4) quickly.

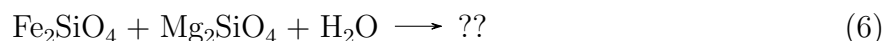


29. (3 points) Let's take a look at the final goal of all these steps: balancing the serpentinization

reaction with any arbitrary value of k .



Note that silica, SiO_2 , is nowhere in this reaction, even though reaction (1) produces it. We'll have to combine reaction (1) with reaction (2) and (4) to obtain a reaction that would be useful for balancing the original reaction:

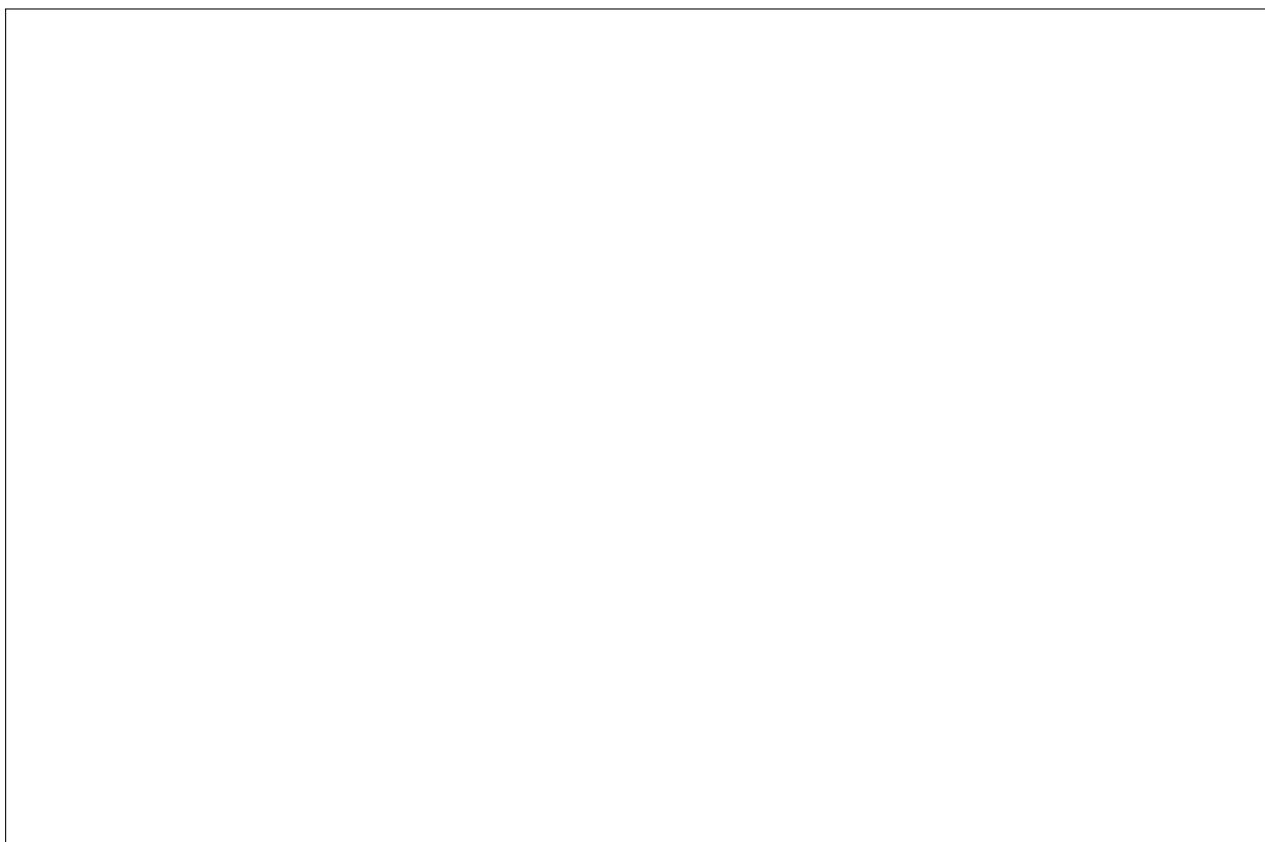
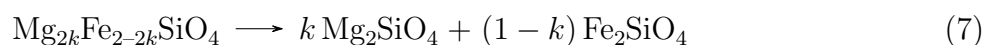


Give the balanced version of reaction (6), where the products are a subset of the products of reaction (5), using the minimum ratio of Mg_2SiO_4 to Fe_2SiO_4 possible.

30. (1 point) One can observe that not all values of k can be successfully balanced by (5) above. For example, take $k = 0$. There is no compounds with Si but without Mg on the product side, while there is Si but no Mg on the reactant side. It turns out that there exists a value ℓ , such that for all $k \geq \ell$, (5) can be balanced using non-negative coefficients. Find ℓ .

31. (3 points) Finally, balance (5) with arbitrary k . For this question only, you may use nonintegral coefficients. You should decompose $\text{Mg}_{2k}\text{Fe}_{2-2k}\text{SiO}_4$ as in (7) and use (2) and (6) to

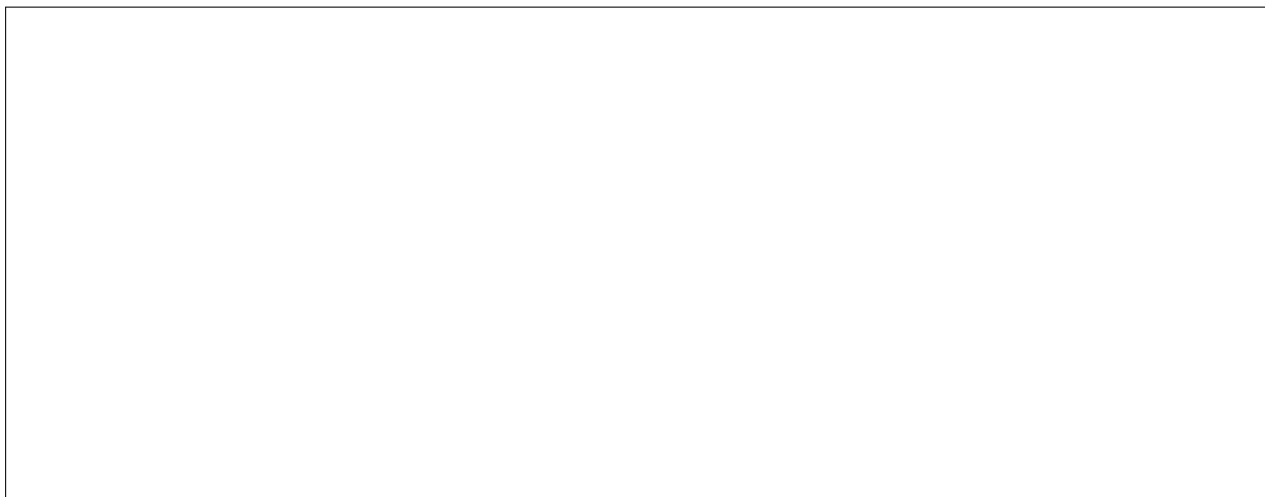
balance the full reaction.



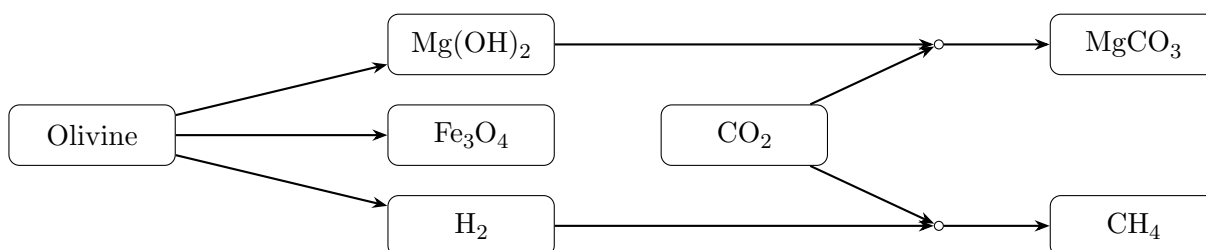
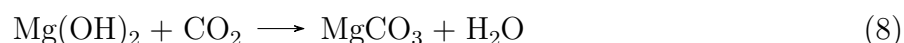
32. (4 points) We'll assume $k = 0.9$ from here to make calculations more concrete. In case you could not fully solve the preceding problems and balance the full reaction, you can use the fact that each molecule of $\text{Mg}_{2k}\text{Fe}_{2-2k}\text{SiO}_4$ will produce $\frac{2-2k}{3}$ molecules of H_2 when hydrolyzed. *This means you can continue from here and still earn points for the rest of the section!*

Some of the H_2 presumably escaped and produced a transient greenhouse effect that allowed liquid water to exist on Mars for an extended period of time.

Calculate the volume of H_2 that is formed per kilogram of $\text{Mg}_{2k}\text{Fe}_{2-2k}\text{SiO}_4$ hydrolyzed at 273 K and 1.5 atm, conditions that would be found on early Mars.



33. (4 points) In 2025, Perseverance reported serpentine-rich rocks at the rim of Jezero Crater and also identified carbonate-bearing ultramafic materials along its traverse. In our simplified model, H_2 produced by serpentinization can (i) escape to the atmosphere, or (ii) react with CO_2 to form CH_4 . Meanwhile, CO_2 -rich water can also react with the $\text{Mg}(\text{OH})_2$ produced during serpentinization to precipitate magnesium carbonate (“magnesite”).



A 1.00 kg sample of $\text{Mg}_{2k}\text{Fe}_{2-2k}\text{SiO}_4$ is fully hydrolyzed, producing H_2 and $\text{Mg}(\text{OH})_2$. The hydrolyzed rock then interacts with 25.0 L of groundwater containing 0.200 M dissolved CO_2 (i.e. $\text{CO}_2(\text{aq})$). In case you could not balance the full reaction above, you may use the assumption that each molecule of $\text{Mg}_{2k}\text{Fe}_{2-2k}\text{SiO}_4$ produces $\frac{6k-3}{8}$ moles of $\text{Mg}(\text{OH})_2$. *Note that this is NOT correct; it merely yields a sensible answer for the following questions. Do not use this value to check your work for reaction (5).*

Find the final concentration of dissolved CO_2 in the 25.0 L groundwater. Partial credit will be given for finding up to 2 of the following: moles of H_2 produced, moles of $\text{Mg}(\text{OH})_2$ produced, total moles of CO_2 remaining after the reactions.

34. (1 point) Find the mass of MgCO_3 formed in the reaction above.

35. (1 point) Find the volume of CH_4 formed in the reaction above.

Ice Cream and Kinetics

Just kidding, this has almost nothing to do with ice cream, although I did think about writing a question involving some really fun calculus based kinetics under changing temperatures, but it turned out to not be doable without a graphing calculator.

What ice cream and this question shares is lactose. In this lab, you will be evaluating the kinetics of lactase supplements.

Instead of using lactase to break down lactose, we're using ONPG as a proxy. It has a shape similar enough to lactose that it is also hydrolyzed by lactase, as we will discuss later.

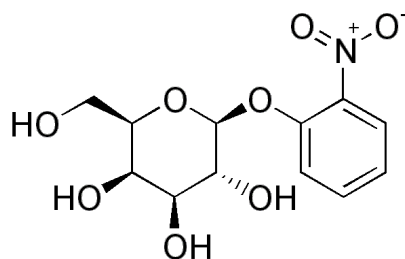
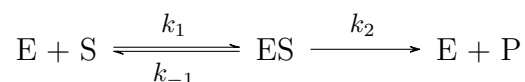


Figure 1: ONPG's chemical structure

36. (4 points) We first rederive the Michaelis–Menten equation from the basic kinetics. Assume the enzyme follows the mechanism



Make the following assumptions:

- $[\text{P}] \approx 0$, so the reverse reaction $\text{E} + \text{P} \rightarrow \text{ES}$ is negligible.
- $\frac{d[\text{ES}]}{dt} \approx 0$ at any given point in time. This is the steady state approximation.
- $[\text{E}]_0 = [\text{E}] + [\text{ES}]$ is constant and represents the amount of enzyme at the start of the reaction.

Show that the initial rate v can be written as

$$v = \frac{V_{\max}[\text{S}]}{K_M + [\text{S}]},$$

and identify $V_{\max}$ and K_M in terms of k_1, k_{-1}, k_2 and $[\text{E}]_0$.

37. (1 point) We will be using the hydrolysis of ortho-nitrophenyl- β -galactoside (ONPG) as the substrate instead of lactose, as it would be easier to measure the hydrolysis of ONPG. When ONPG is hydrolyzed, it produces ortho-nitrophenol, which is yellow in color and has an absorbance peak that can be measured at 420 nm.

Hypothetically, a different chromogenic substrate is cleaved to form a product that absorbs strongly at 700 nm. What color would that product solution appear as?

38. (15 points) Now, let's determine V_{max} and K_M for our reaction.

The grading of this question is somewhat complicated so we'll explain it real quick. *Feel free to skip to the actual lab, just do your best!* Up to 3 points will be given for data collection and presentation; maximum score will be given for plotting $1/v$ against $1/[S]$, where $v = \frac{d[P]}{dt}$, along with correctly labeled axis and a line of best fit. An additional 6 points will be given for

the accuracy of V_{max} , and similarly 6 points will be given for the accuracy of K_M . For each of the accuracy based measures, the top team will score 6 points, and the median team will score 2 point. This will be used to create a linear fit for the rest of the teams. Teams cannot score below 0 points for either subpart.

Given constants:

- Molar mass of ONPG: 301.25 g/mol.
- Beer-Lambert law: $A = \varepsilon \ell c$.
- $\varepsilon_{420} = 4.50 \times 10^3 \text{ L mol}^{-1} \text{ cm}^{-1}$, and $\ell = 1.00 \text{ cm}$.
- ONPG stock: 1.00 mg/mL.

Suggested protocol

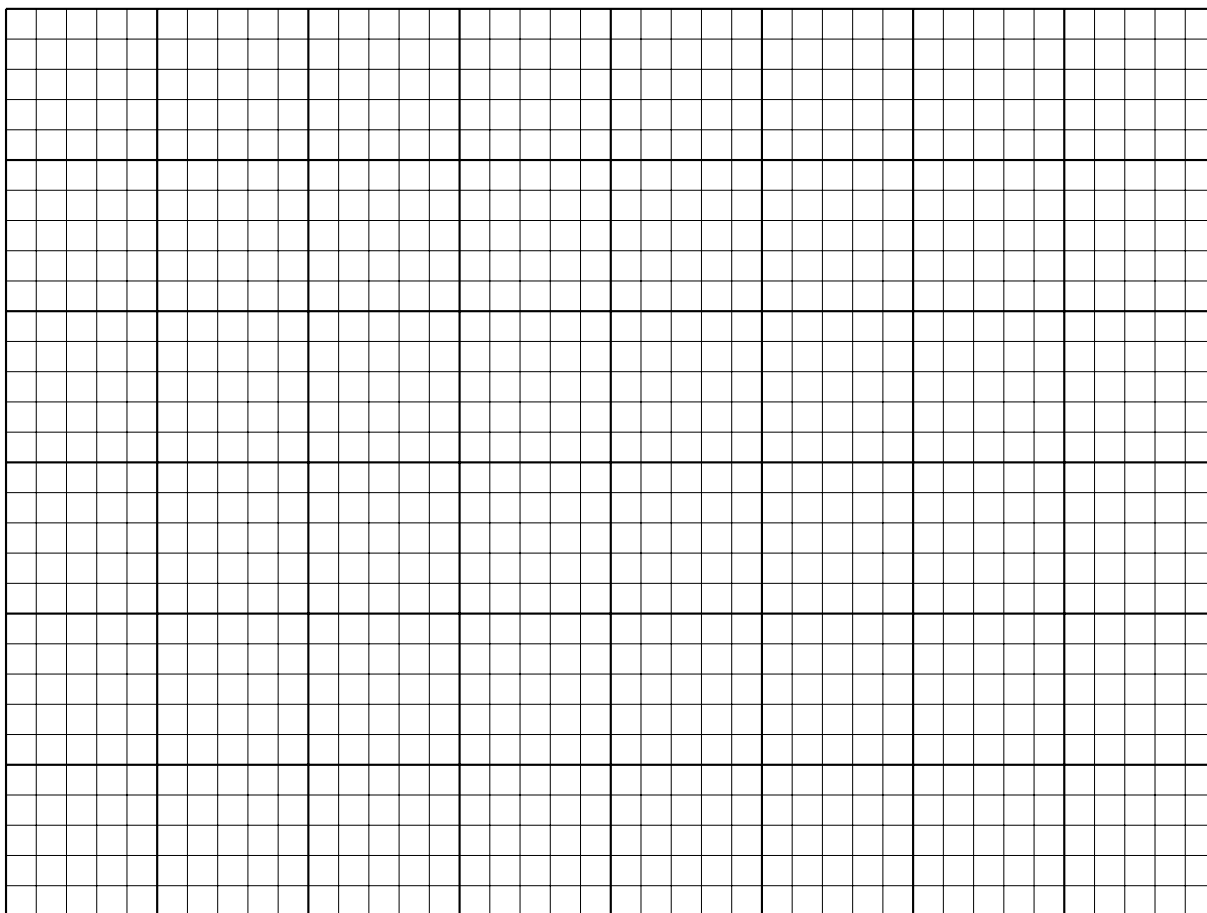
- For each concentration of ONPG you decide to use (suggestions will be given on the board), perform the following:
- Prepare a 2.00 mL reaction in a cuvette by mixing ONPG solution + buffer/water to a total of 1500 μL .
- Start the reaction by adding 500 μL enzyme stock, quickly mix, and immediately begin timing.
- Record A_{420} at $t = 0, 60 \text{ s}$.
- Compute the initial slope $\frac{\Delta A}{\Delta t}$ (in absorbance/min), then convert to rate v (in $\mu\text{M}/\text{min}$).

[S] (mM)	$A_{420}(0 \text{ s})$	$A_{420}(60 \text{ s})$	$[\text{S}]^{-1}$ (x axis)	v^{-1} (y axis)

To determine the values for $V_{\max}$ and K_M , we can use the Lineweaver-Burk plot:

$$\frac{1}{v} = \frac{K_M}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}.$$

Thus, when you plot $y = \frac{1}{v}$ against $x = \frac{1}{[S]}$, the **y-intercept** is $\frac{1}{V_{\max}}$ (so $V_{\max} = 1/(\text{y-int})$) and the **slope** is $\frac{K_M}{V_{\max}}$ (so $K_M = (\text{slope}) \cdot V_{\max}$).



Calculate the slope and y-intercept, and give your final values for $V_{\max}$ and K_M .

39. (5 points) Two common inhibitor models can be distinguished by their effects on Michaelis–Menten parameters:

- Competitive: $V_{\max}$ unchanged, apparent K_M increases.
- Noncompetitive: apparent $V_{\max}$ decreases, K_M unchanged.

You are also provided with a solution of an inhibitor labeled X . Design and carry out an experiment to conclude whether or not it is competitive. You may use the following graph paper as necessary. It is suggested to use 500 μL of the inhibitor for a 2.00 mL reaction to obtain optimal results.

